

TRANSACTIONAL ANALYTICS

CUSTOMER SHOPPING BEHAVIOR

What 3,900 transactions reveal about retention, discounts, product focus, and the next analytics tests.

DATASET VOLUME

3,900

Verified purchases

METRIC FIELDS

18

Transaction attributes

DECISION AGENDA



Focus: convert repeat volume into an owned relationship.

THE GROWTH PROBLEM IS CONVERSION TO RELATIONSHIP

Large repeat base

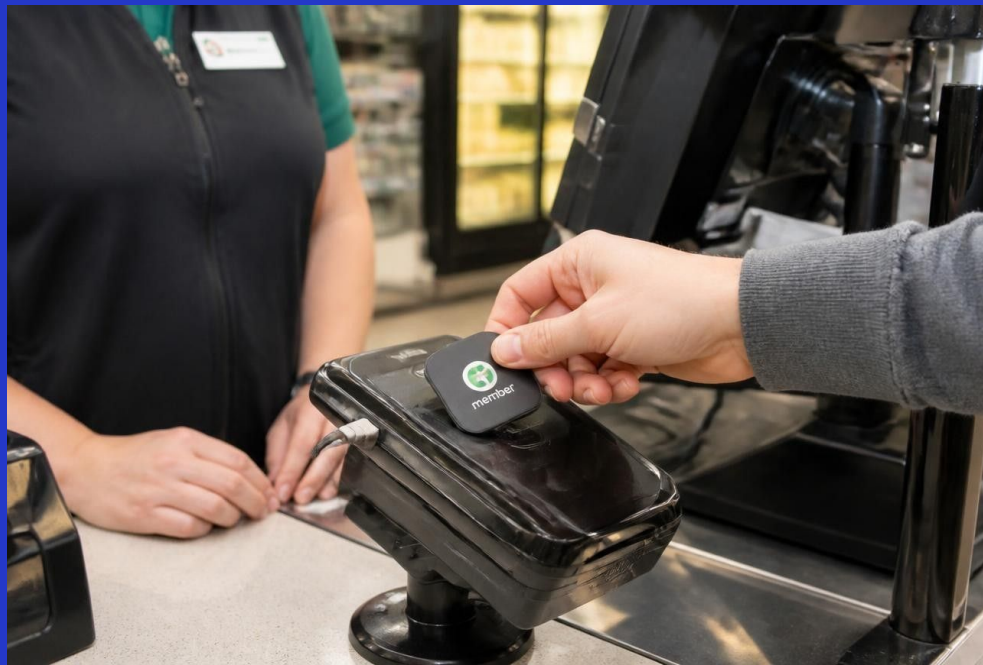
3,116 customers are classified as Loyal, while only 1,053 customers are active subscribers.

The spend parity paradox

Non-subscribers generate more total revenue because they are the larger pool (\$170,436 vs \$62,645), but average spend is nearly identical: \$59.87 vs. \$59.49 for subscribers.

Strategic implication

The opportunity is not simply to acquire more customers; it is to convert proven repeat behavior into an owned relationship.



WHAT THE DATASET CAN AND CANNOT ANSWER

01

Raw transactions

3,900 purchases · 18 fields

Covering demographics, products, purchase context, shipping, promotions, reviews, and subscription status.

02

Cleaned table

Data preparation

Review-rating gaps were imputed with the median rating by product category; promo-code-used was dropped as redundant with discount-applied.

03

SQL analysis

Feature engineering

Age groups and purchase-frequency days were engineered.

04

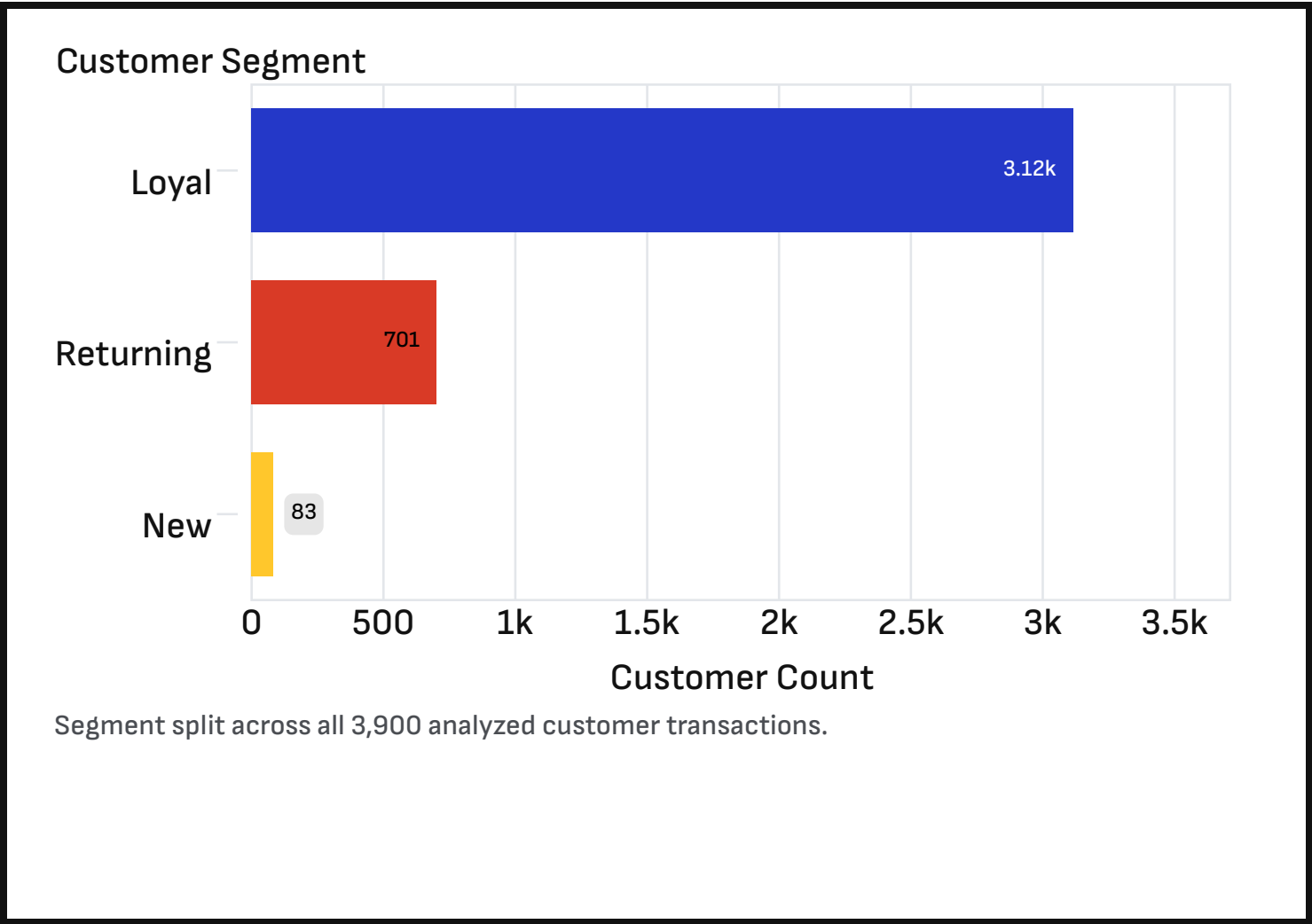
Decision tests

Read carefully

It does not prove that subscriptions, discounts, or shipping caused higher spend.

Analytical boundary: This is transactional analysis, so patterns are directional.

REPEAT BEHAVIOR IS ALREADY THE DOMINANT SIGNAL



STRATEGIC TAKEAWAY

80%

Of all buyers are already classified as Loyal.

New customers represent just 83 buyers. Build the next growth program around moving 701 returning buyers into loyalty, then layer subscription benefits on top.

SUBSCRIPTION IS UNDER-PENETRATED, NOT UNDER-SPENDING

Subscriber Status	Customer Count	Average Spend	Total Revenue
Subscribers (Yes)	1,053 (27%)	\$59.49	\$62,645
Non-Subscribers (No)	2,847 (73%)	\$59.87	\$170,436

Observational revenue and spend split by subscription flag across 3,900 records.

ECONOMIC INSIGHT

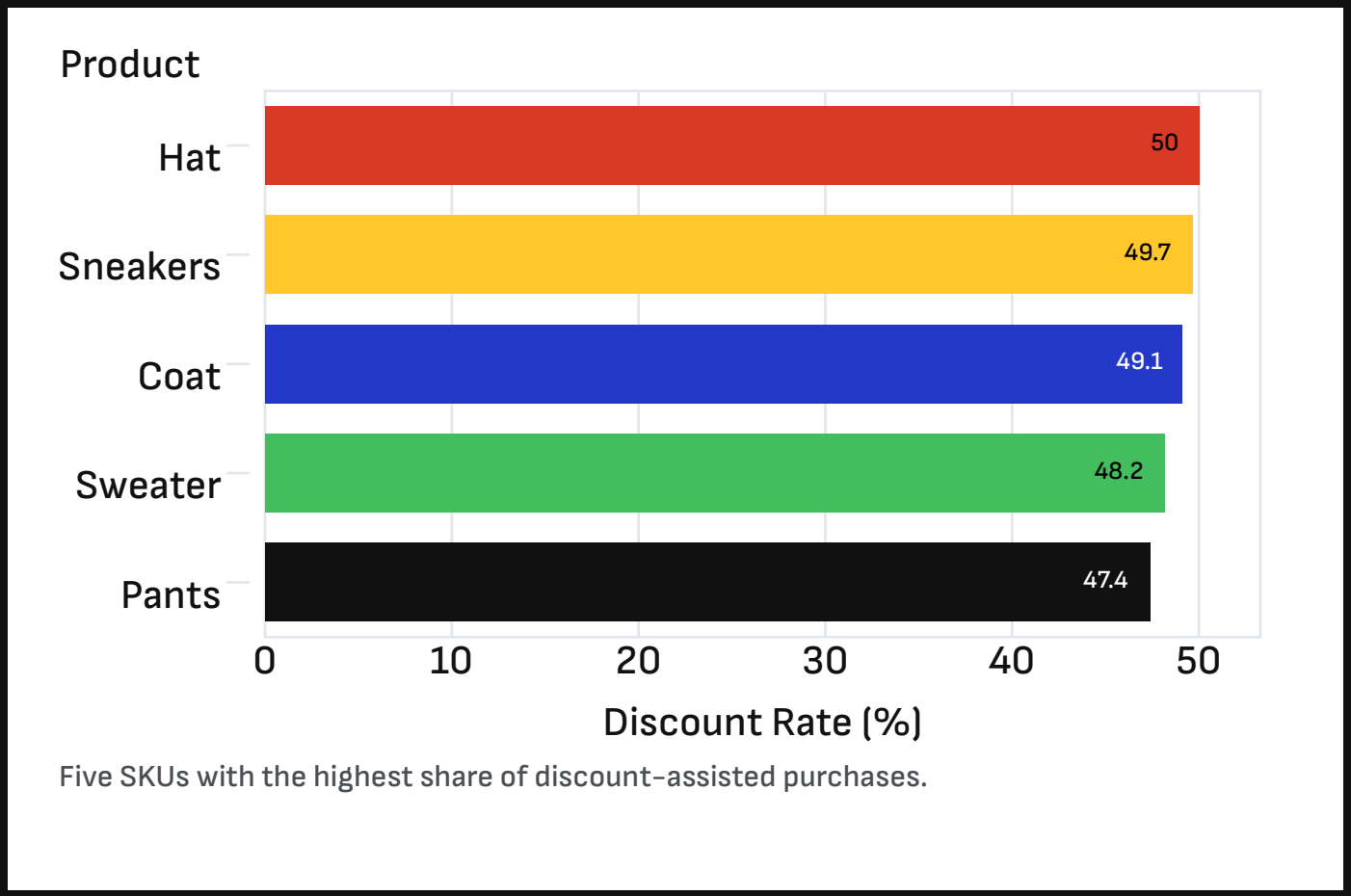
-\$0.38

Spend gap between cohorts is negligible.

Testable hypothesis: sell subscription as convenience, recognition, or access rather than as a discount-only proposition.

PROMOTION DISCIPLINE

DISCOUNTS ARE CONCENTRATED ENOUGH TO MANAGE SURGICALLY



MARGIN PRESERVATION



839 BUYERS

839 customers used discounts yet spent above the \$59.76 benchmark. Shift away from site-wide sales to category-specific margin holdout tests.

PRODUCT SIGNALS POINT TO TWO DIFFERENT JOBS

PROOF PRODUCTS • BY RATING

Gloves: 3.86

Sandals: 3.84

Boots: 3.82

Hat: 3.80

Skirt: 3.78

TRAFFIC DRIVERS • BY VOLUME

Jewelry: 171 orders (Accessories)

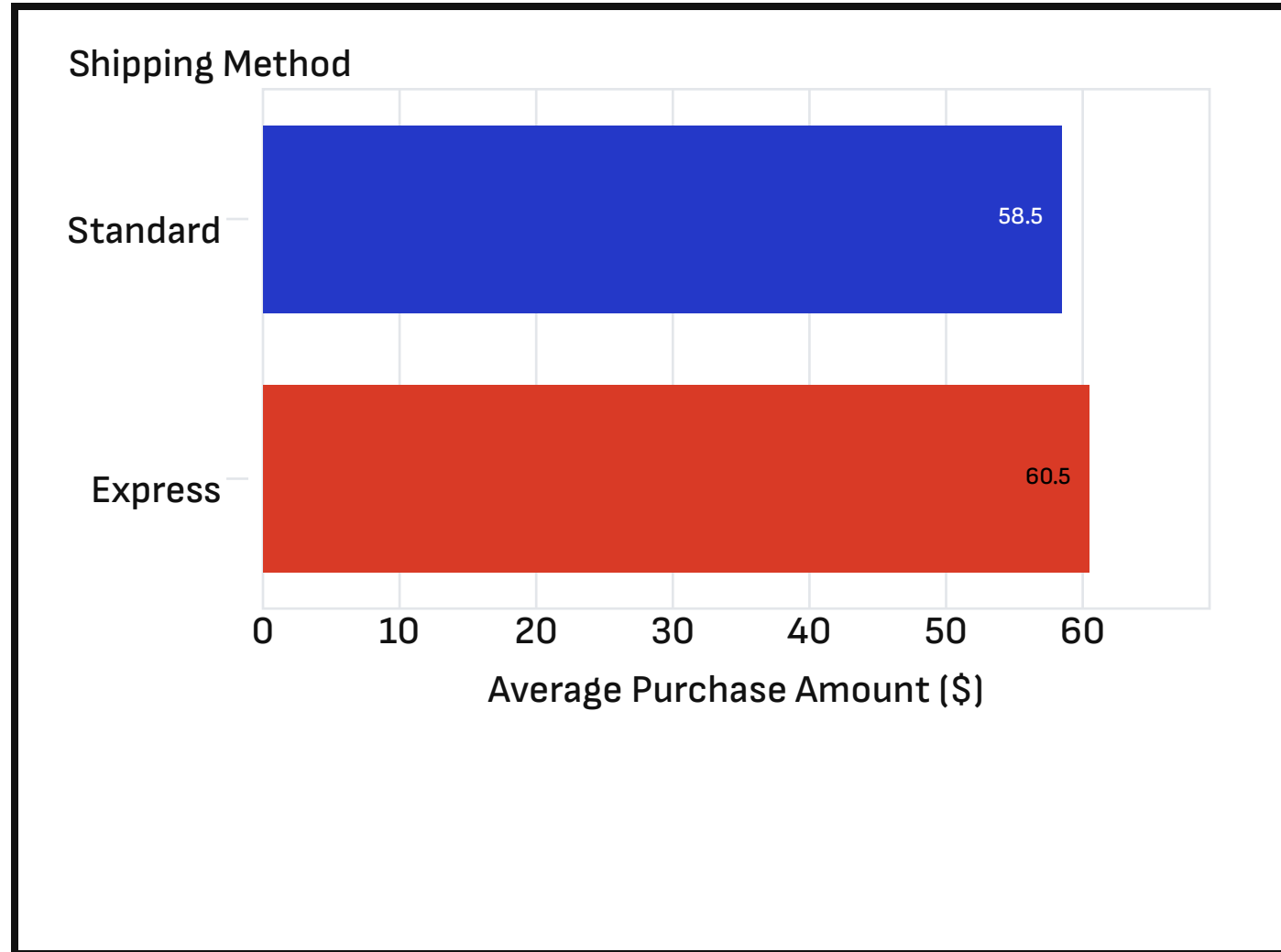
Blouse and Pants: 171 orders each (Clothing)

Sandals: 160 orders (Footwear)

Jacket: 163 orders (Outerwear)

Decision: use top-rated products as proof and best sellers as traffic drivers; do not assume the same products should play both roles.

FASTER SHIPPING EARNS A MEASURABLE PREMIUM



TEST HYPOTHESIS

+\$2.02

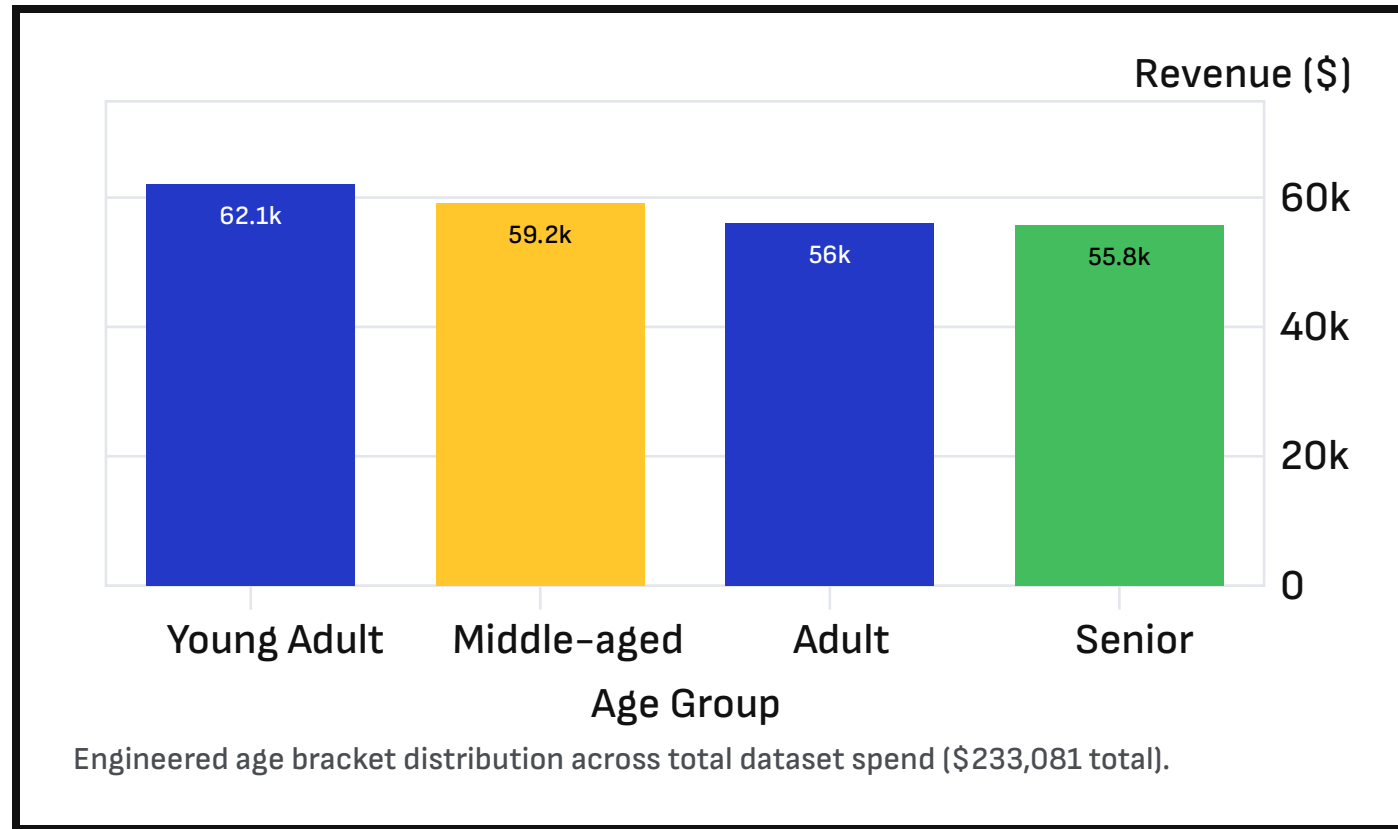
Average purchase amount difference.

Faster delivery aligns with slightly higher baskets in this dataset, but the relationship is not causal.

Next test: Pair express eligibility with high-intent or high-value journeys and measure conversion, basket size, and contribution margin together.

DEMOGRAPHIC BREAKDOWN

YOUNG ADULTS ARE THE LARGEST AGE-GROUP REVENUE POOL



AUDIENCE ALLOCATION

\$62.1K

Young Adults lead, but the spread remains balanced.

Young Adults generate \$62,143, while Seniors represent \$55,763. Lead acquisition and offer testing with Young Adults; tailor creative and merchandising to remaining groups rather than collapsing them into one audience.

TURN THE FINDINGS INTO A 90-DAY ANALYTICS AGENDA

01

Convert loyalty into subscription

Target returning and loyal customers with benefit-led messages.

Track: Subscription conversion and repeat purchase rate.

02

Control discount leakage

Run holdouts on Hat, Sneakers, Coat, Sweater, and Pants.

Track: Incremental units, revenue, and margin.

03

Use proof and traffic differently

Feature top-rated products in trust-building creative and best sellers in demand capture.

Track: Click-through and conversion by role.

04

Test shipping as a value lever

Target express offers to high-intent journeys.

Track: Basket size, conversion, fulfillment cost, and margin.